

Artifact Description/Artifact Evaluation: A Reproducibility Bane or a Boon

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Artifact Description/Artifact Evaluation

A process to establish reproducibility of computational artifacts.

Conducted from the point of manuscript submission to its acceptance at a conference or a journal.



Artifact Description/Artifact Evaluation

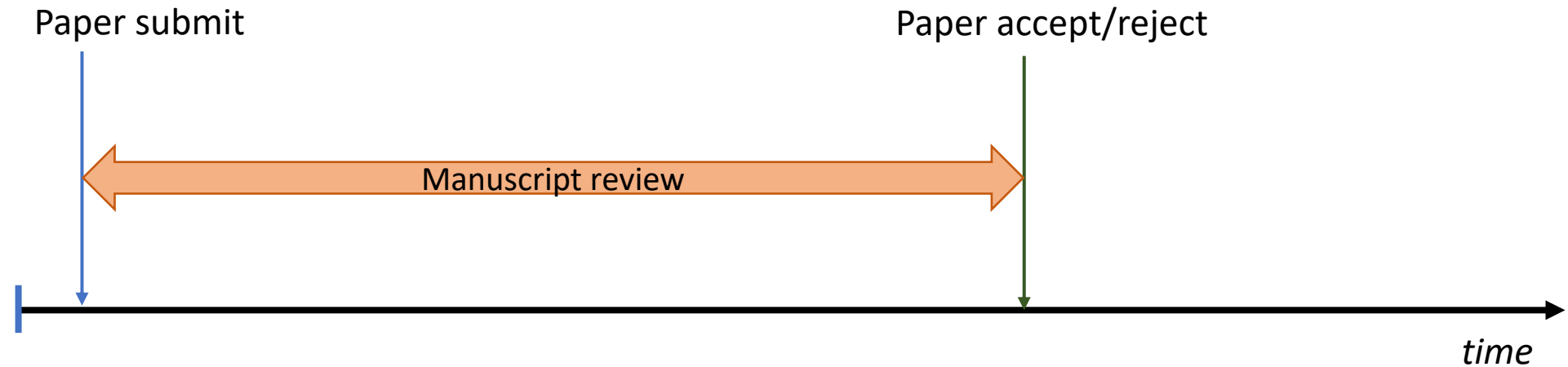
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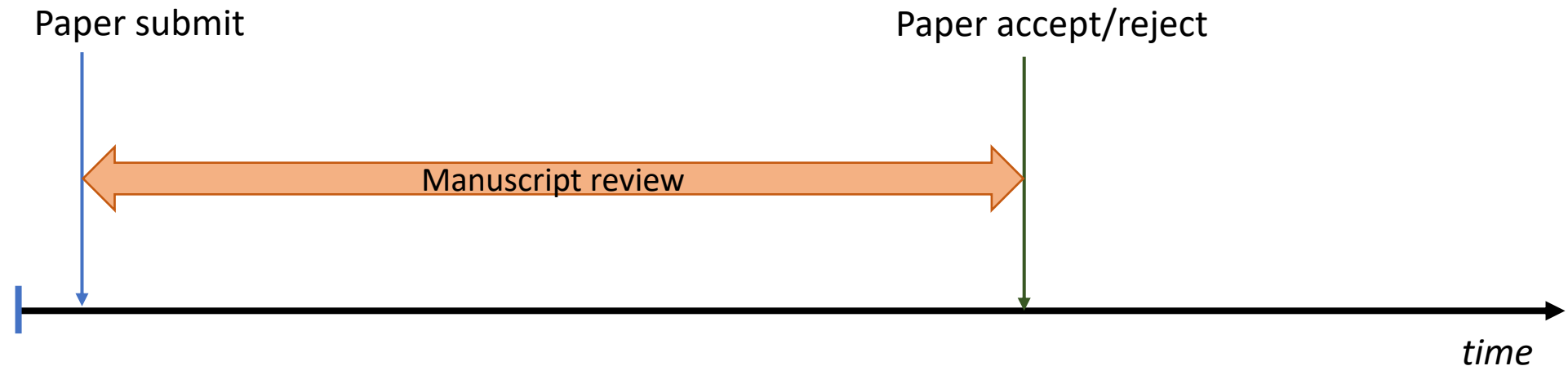
OSDI	EuroSys	Supercomputing	Middleware	VLDB	CGO	MICRO	PoPPL
SOSP	FAST	PACT	SIGMOD	ASPLOS	MLSys	NeuroIPS	



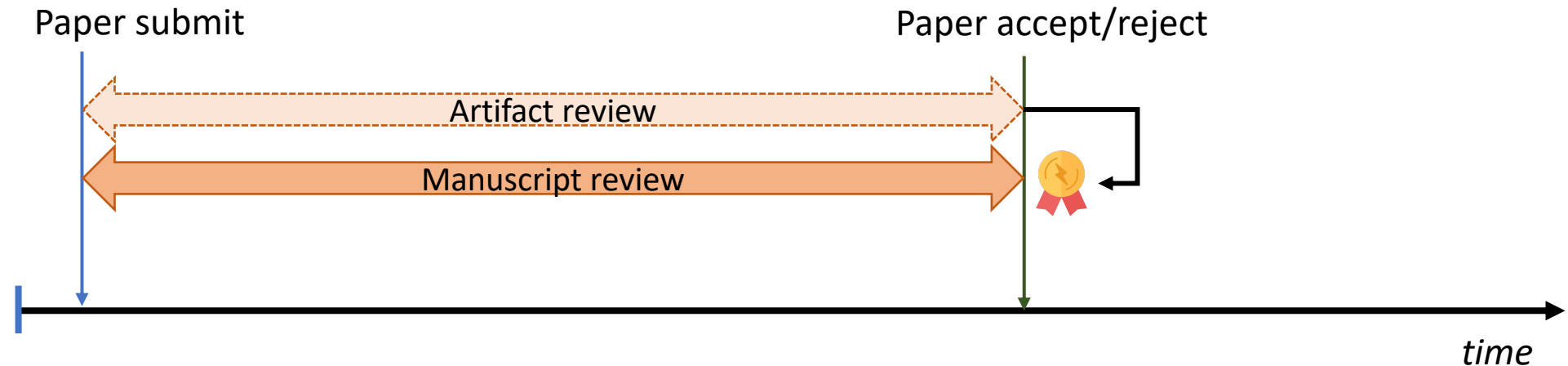
Manuscript review



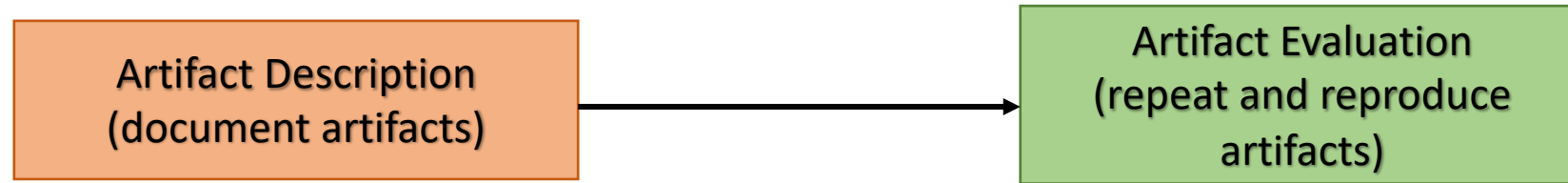
Manuscript + Artifact review



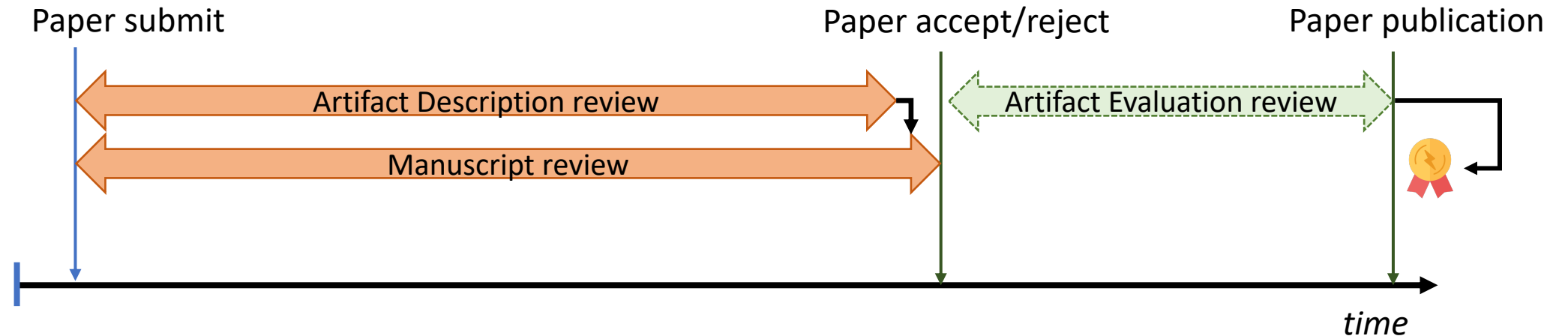
Manuscript + Artifact review



Manuscript + Artifact Description/Artifact Evaluation review



Manuscript + Artifact Description/Artifact Evaluation review



SC AD/AE Historical Evolution

- Source: SC21 website

Year	Artifact Description (AD)	Artifact Evaluation (AE)
2015	Optional	
2016	Optional	
2017	Optional	
2018	Optional	
2019	Mandatory	Optional
2020	Mandatory	Submit artifacts with badges
2021	Mandatory	Submit artifacts with badges



Does AD/AE improve participation?

- Source: SC21 website

Year	Artifact Description (AD)	Artifact Evaluation (AE)	# of Papers (AD)	# of Papers (AE Requested)
2015	Optional		1	1
2016	Optional		9	3
2017	Optional		39	-
2018	Optional		-	-
2019	Mandatory	Optional	-	-
2020	Mandatory	Submit artifacts with badges	379 (96.4%)	112 (30%)
2021	Mandatory	Submit artifacts with badges	336 (92.8%)	188 (50%)

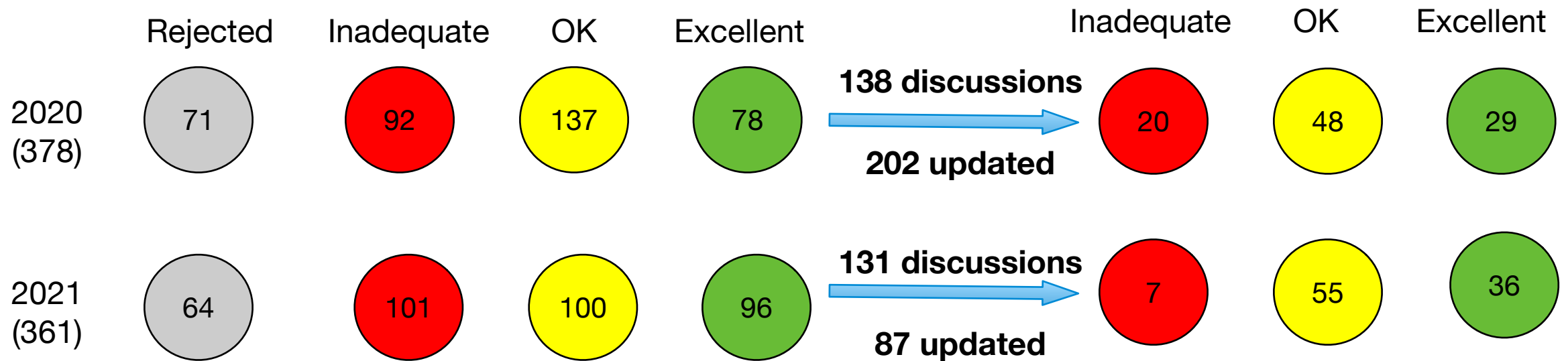


Does participation translate to improved reproducibility?



Does AD improve reproducibility?

- Source: Supercomputing Linklings submission site
- SC20-21 AD/AE Chairs



Mere participation does not translate to improved reproducibility



Does AE improve reproducibility?

- Source: Andrew Young, Sandia National Laboratory
- SC20-21 AD/AE Chairs

2020 Of the 97 accepted papers
 37 had artifacts published and ACM badges awarded

2021 Of the 99 accepted papers
 ? had artifacts published and ACM badges awarded

Focus on improvements rather than punishments
Demonstrates commitment to educational experience

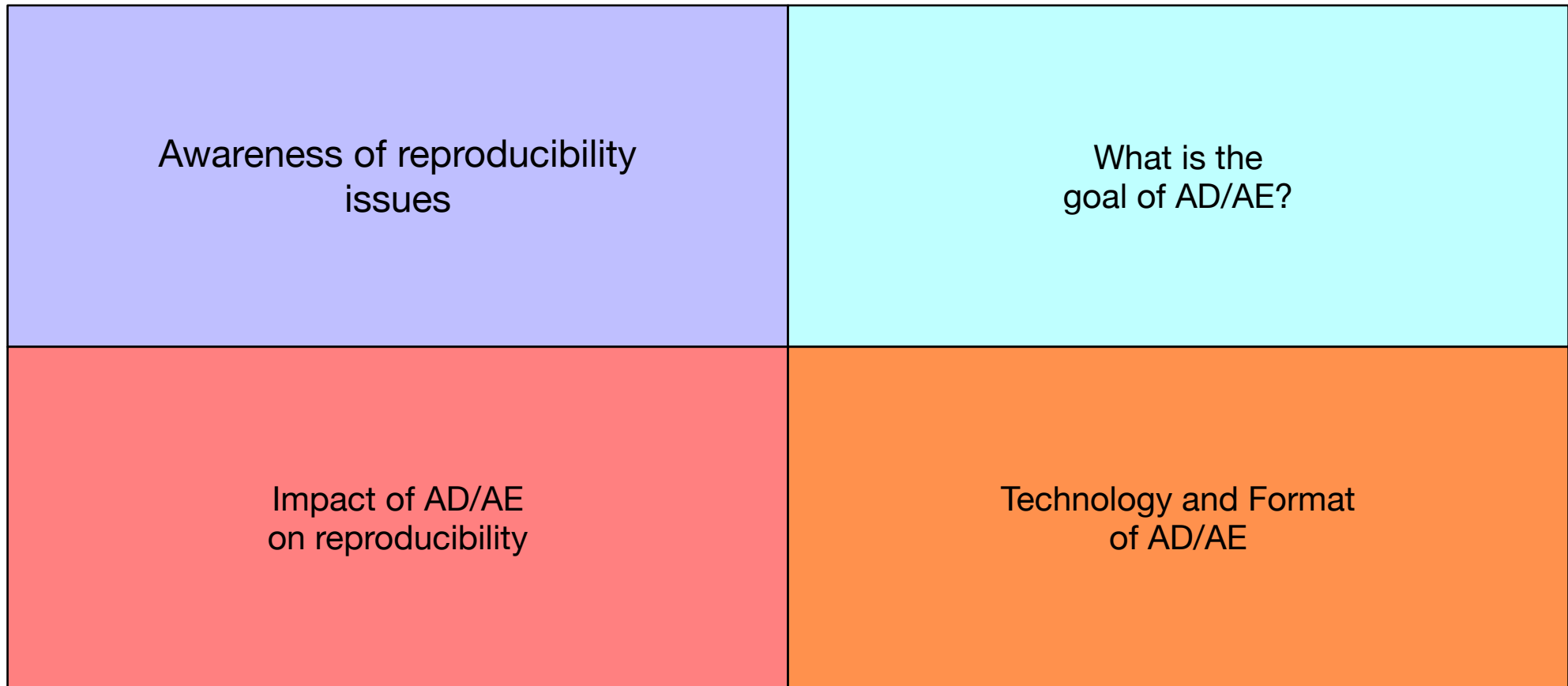


User Feedback

- Source: *Transparency and Reproducibility Practice in High Performance Computing*. Plale, Malik, Pouchard. Forthcoming in IEEE Computer, 2021.
- Conducted in Aug '20
- 9,949 unique recipients who had participated in the SC17-19 technical programs
- 204 self-selected participants responded to the survey.

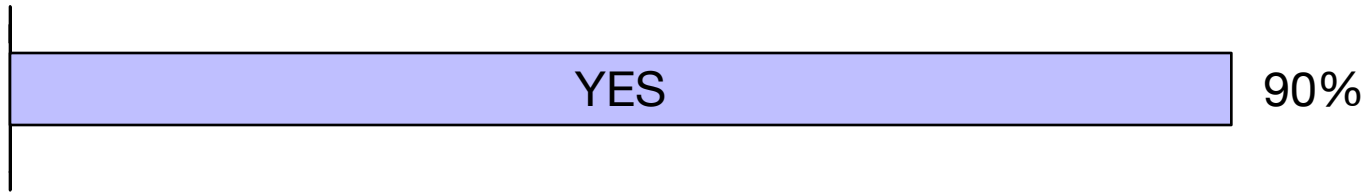


Survey Categories

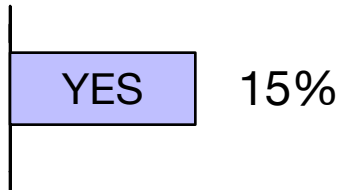


Awareness of Reproducibility Issues

Is there a reproducibility crisis in computational science and engineering?

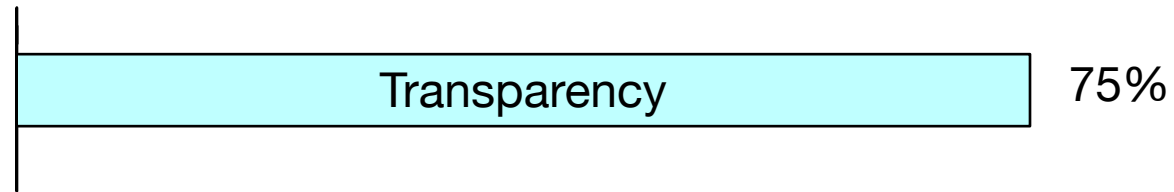


Are concerns about reproducibility exaggerated?

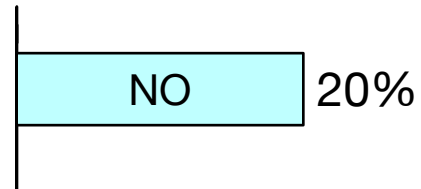


Goal of AD/AE: Transparency Vs Reproducibility

Transparency Vs Reproducibility: Which is the goal?

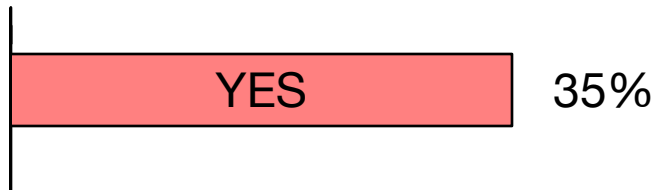


Do you know the difference between transparency and reproducibility?

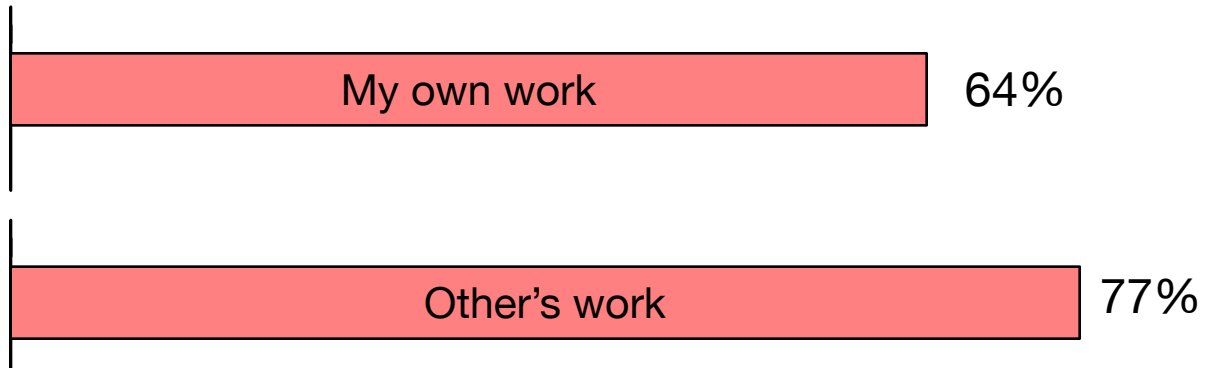


Impact of AD/AE on reproducibility

Have you ever used Appendix information from other SC papers?

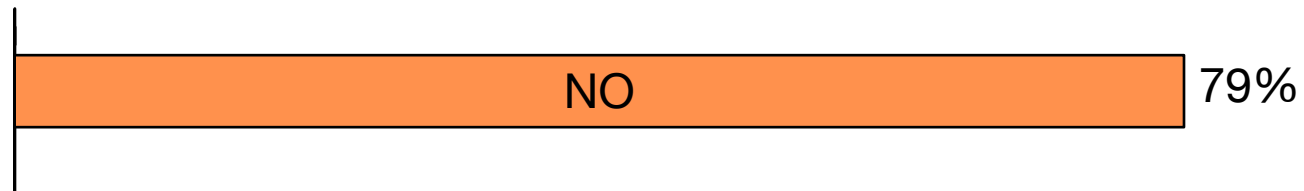


Does AD/AE help you think about reproducibility of

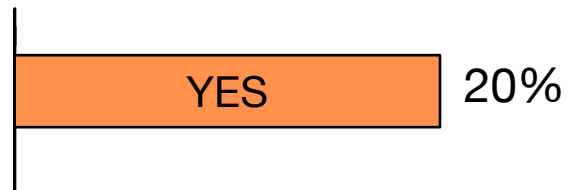


Technology for AD/AE

It is sufficient to publish code in a repository for the purpose of reproducibility



Did you use containers to share results?



Cost of AD/AE: High time investment

- Author time:
 - For majority limited to AD but can be a 1-2 worth of work.
 - For AE it is much longer but currently allocated as a maximum of 6 weeks.
- Reviewer time:
 - 4-5 hours for AD
 - 20 hours for AE with 10 hours per artifact.



Where does author time go?

- Improving quality code
- Improving design
- Documenting environment and files
- Testing artifacts



Where does reviewer time go?

- Identifying files for an application is a challenge
- Missing workflows
 - Really, that data/algorithm should be part of the bundle?
 - No tools to check if everything relevant for the publication is submitted
- No mapping of experiments to content in the paper.
 - No available infrastructure for efficiently verifying claimed results

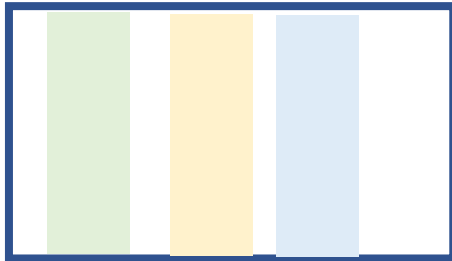
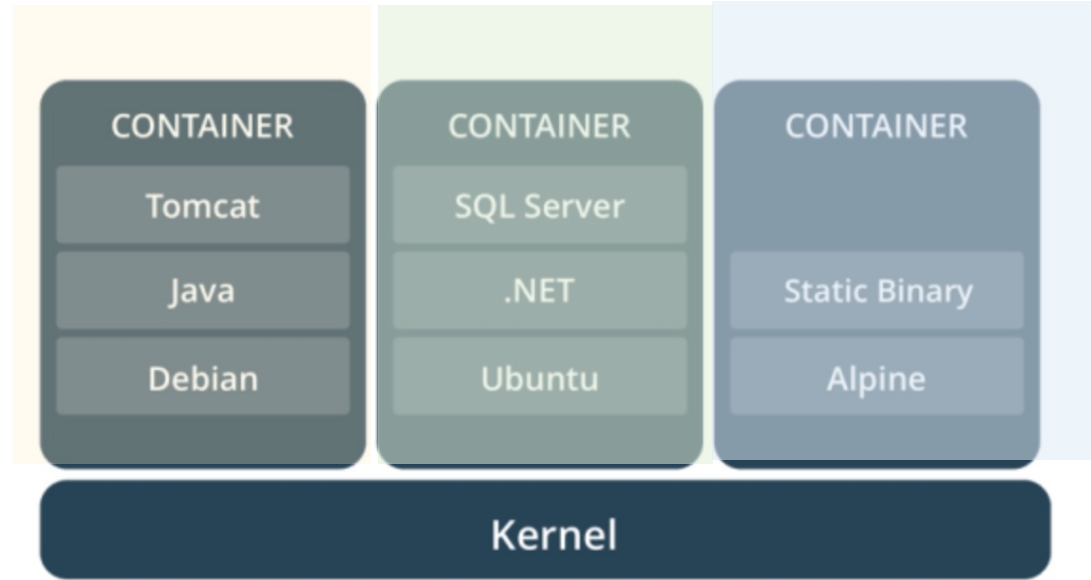


Systems for reproducible research

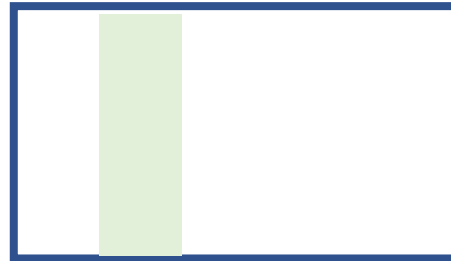


For authors

Containers provide constrained resource isolation



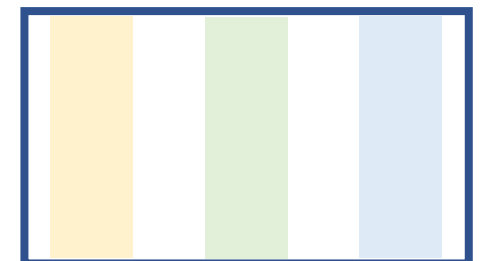
CPU



Memory



Filesystem



Network





Docker: Using containers from build to run



Build

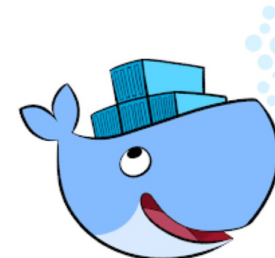
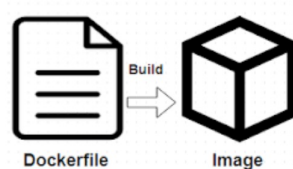


Ship



Run

- Build images that captures applications requirements.
- Manually commit or use a recipe file.
- Push an image to DockerHub, a hosted registry, or a private Docker Registry.
- Share Images
- Use Docker Engine to pull images down and execute a container from the image.



<https://www.exascaleproject.org/event/conthpc>



Authors must program a Dockerfile

```
1  # fetch node v4 LTS codename argon
2  FROM node:argon
3
4  # Request samplename build argument
5  ARG samplename
6
7  # Create app directory
8  RUN mkdir -p /usr/src/spfx-samples
9  WORKDIR /usr/src/spfx-samples
10
11 #Install app dependencies
12 RUN git clone https://github.com/SharePoint/sp-dev-fx-webparts.git .
13 WORKDIR /usr/src/spfx-samples/samples/$samplename
14
15 # install gulp on a global scope
16 RUN npm install gulp -g
17
18 # RUN ["npm", "install", "gulp"]
19 RUN npm install
20 RUN npm cache clean
21
22 # Expose required ports
23 EXPOSE 4321 35729 5432
24
25 # Run sample
26 CMD ["gulp", "serve"]
27
```

```
FROM buildkit-export AS buildkit-buildkitd.oci_only
COPY --from=buildkitd.oci_only /usr/bin/buildkitd.oci_only /usr/bin/
COPY --from=buildctl /usr/bin/buildctl /usr/bin/
ENTRYPOINT ["buildkitd.oci_only"]

# Copy together all binaries for containerd worker mode
FROM buildkit-export AS buildkit-buildkitd.containerd_only
COPY --from=runc /usr/bin/runc /usr/bin/
COPY --from=buildkitd.containerd_only /usr/bin/buildkitd.containerd_only /usr/bin/
COPY --from=buildctl /usr/bin/buildctl /usr/bin/
ENTRYPOINT ["buildkitd.containerd_only"]

FROM alpine AS containerd-runtime
COPY --from=runc /usr/bin/runc /usr/bin/
COPY --from=containerd /go/src/github.com/containerd/containerd/bin/containerd* /usr/bin/
COPY --from=containerd /go/src/github.com/containerd/containerd/bin/ctr /usr/bin/
VOLUME /var/lib/containerd
VOLUME /run/containerd
ENTRYPOINT ["containerd"]

FROM buildkit-${BUILDKIT_TARGET}
```

Containers do not reduce time for conducting reproducible research



- Declarative encapsulation of dependencies for isolated execution
 - E.g. various shell utilities and library versions unknown to user





Automatic Encapsulation of Dependencies: The Sciunit





Key Idea: Identify dependencies during program execution

- Captures application dependencies during executions
- Repeats executions (with guarantees) within isolated environments



Sciunit steps and external requirements



1. Create



C/C++



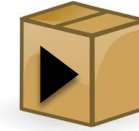
julia



jupyter



2. Share



3. Repeat



CloudLab





Use Sciunit for your next paper submission!

1. Tools downloaded ~1150 times (tracked using pip)
2. 10 active contributors to the project
3. Actively used in geoscience disciplines that develop computational models and data-analytic pipelines

Website: <http://sciunit.run>

Issues and contribution: pr@sciunit.run



For reviewers

Reviewer Task

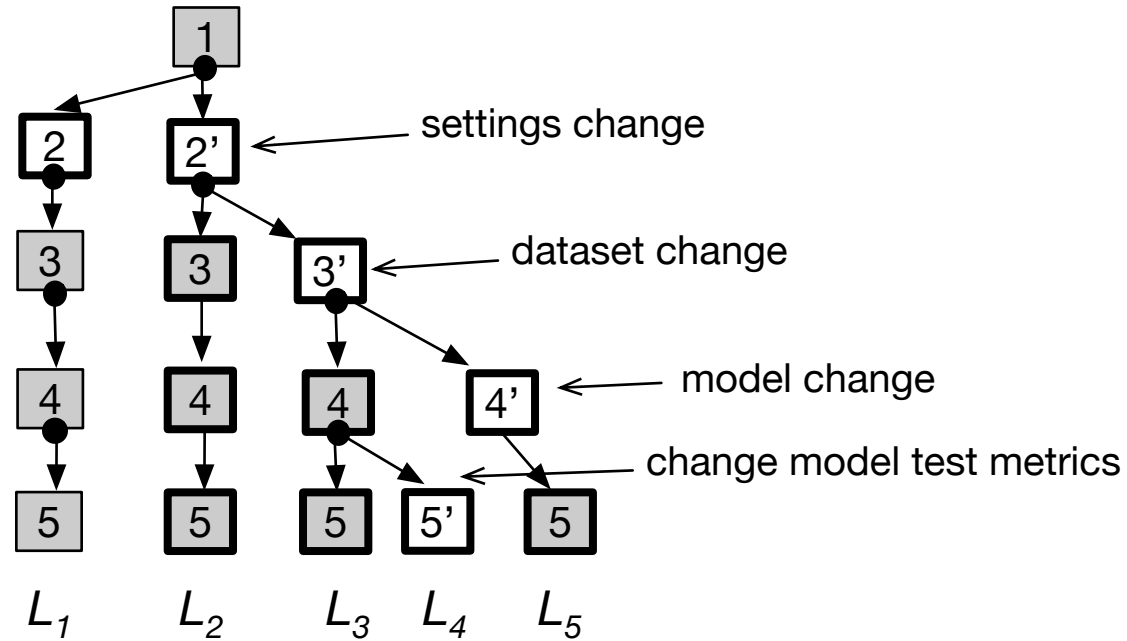
(a) **Kuzushiji**

Model	Loss	Top-1	Top-3	Top-5	R	# p
ResNet-18	0.0709	98.23	99.53	99.79	100	11.1M
ResNet-18+DeepCompression	0.1617	95.92	98.91	99.58	51.49	5.7M
ResNet-18+EDropout(F)	0.1112	97.75	99.41	99.78	100	11.1M
ResNet-18+EDropout(P)	0.1107	97.78	99.45	99.73	51.49	5.7M
ResNet-34	0.0704	99.52	99.90	99.93	100	21.2M
ResNet-34+DeepCompression	0.2023	94.42	98.66	99.55	46.11	9.8M
ResNet-34+EDropout(F)	0.1115	97.78	99.42	99.72	100	21.2M
ResNet-34+EDropout(P)	0.1143	97.71	99.44	99.65	46.11	9.8M
ResNet-50	0.0902	97.70	99.44	99.79	100	23.5M
ResNet-50+DeepCompression	0.2142	94.36	98.53	99.36	45.57	10.7M
ResNet-50+EDropout(F)	0.1250	97.89	99.38	99.75	100	23.5M
ResNet-50+EDropout(P)	0.1289	97.63	99.35	99.73	45.57	10.7M
ResNet-101	0.0699	98.26	99.63	99.80	100	42.5M
ResNet-101+DeepCompression	0.5648	93.30	98.48	99.39	46.00	19.5M
ResNet-101+EDropout(F)	0.1140	98.06	99.39	99.69	100	42.5M
ResNet-101+EDropout(P)	0.1087	98.05	99.47	99.69	46.00	19.5M
AlexNet	0.1162	97.77	99.43	99.85	100	57M
AlexNet+EDropout(F)	0.1976	96.53	99.25	99.72	100	57M
AlexNet+EDropout(P)	0.2089	96.57	99.24	99.68	78.57	44.8M
SqueezeNet	0.2114	97.19	99.27	99.72	100	0.72M
SqueezeNet+EDropout(F)	0.2414	96.45	99.05	99.57	100	0.72M
SqueezeNet+EDropout(P)	0.2411	96.35	98.91	99.54	49.86	0.36M

- Source: “EDropout: Energy-Based Dropout and Pruning of Deep Neural Networks”, [Hojjat Salehinejad](#), [Shahrokh Valaee](#), In [IEEE Transactions on Neural Networks and Learning Systems](#), 2020.
- Time to run each model is: 22 minutes.
- Number of versions: 22
- Can we reduce reviewer time?



CHEX: Multiversion Execution

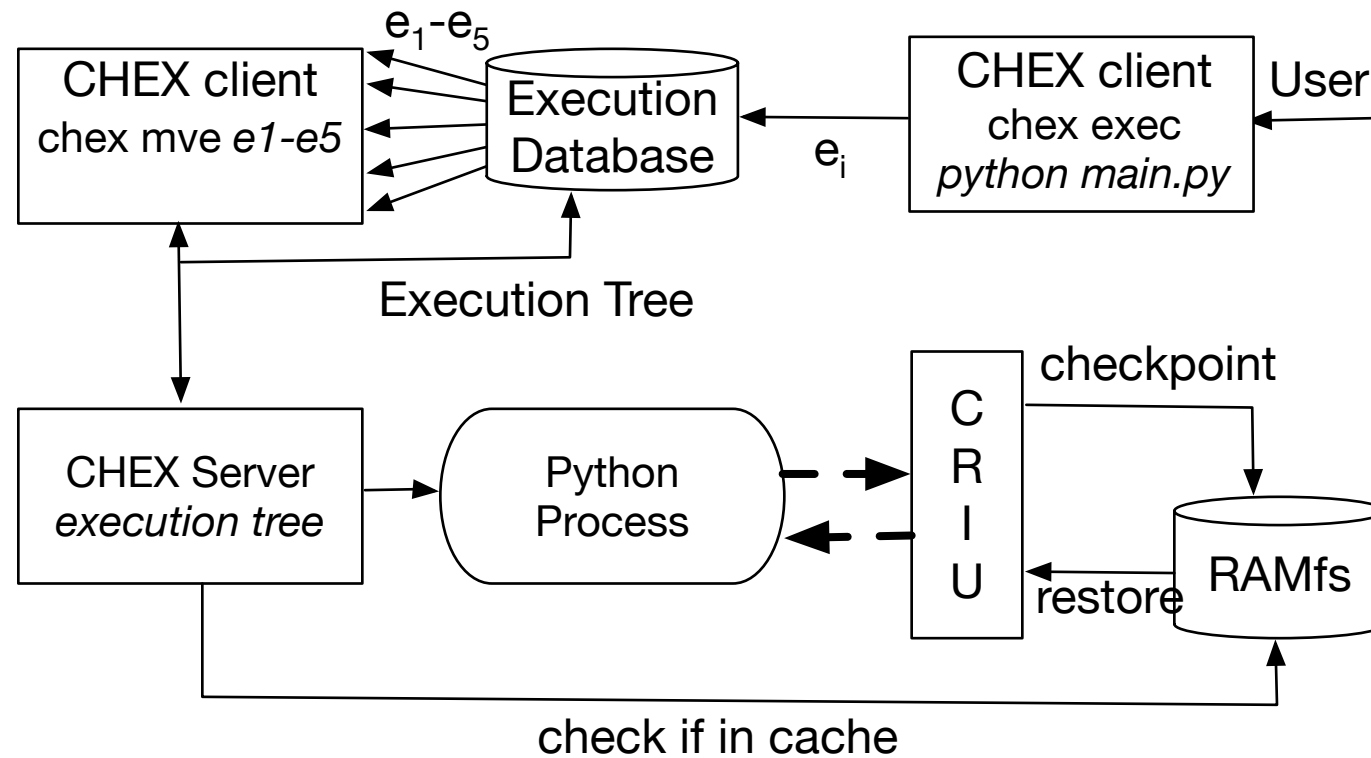


$(C_1, c=1, s=1)$
 $(C_2, c=1, s=2)$
 $(C_2', c=1, s=2)$
 $(C_3, c=4, s=20)$
 $(C_3', c=2, s=20)$
 $(C_4, c=20, s=22)$
 $(C_4', c=10, s=22)$
 $(C_5, c=2, s=24)$
 $(C_5', c=2, s=24)$

- i' changed cell content
- i changed program state
- changed program state

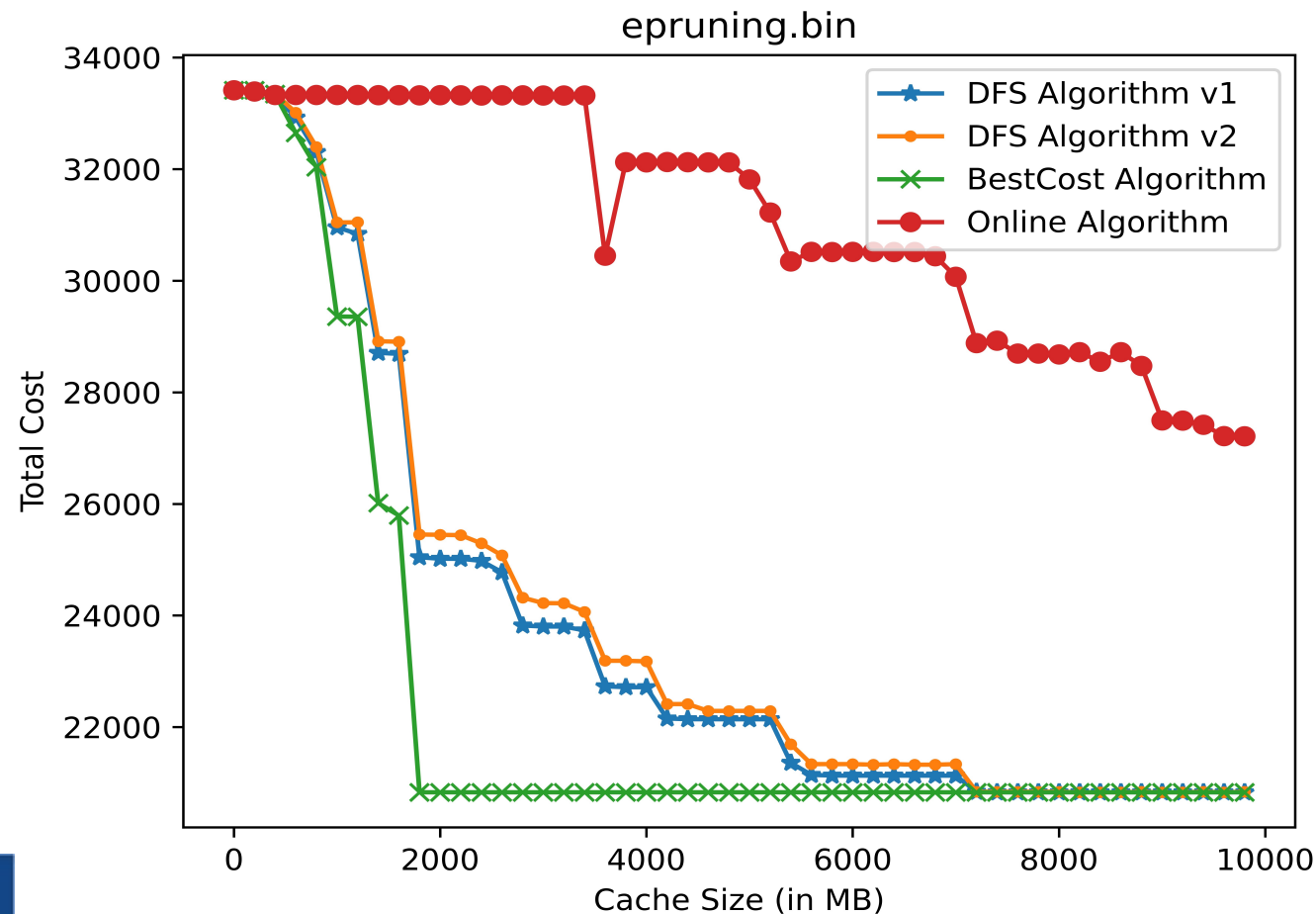


CHEX: Creates Smarter Replay Orders



N. Manne, T. Malik, A. Chaudhary, A. Bagchi. CHEX: Multi-version Execution with Ordered Checkpoints. *Submitted, 2021*

Reduction in application runtime



25 versions

9-13 branches

Total compute time:
33500

Total storage size: 57
GB



Closing Remarks

- Reproducibility is achievable with careful attention to process
 - Encouraging authors
 - Creating an educational and learning experience
- Time is a constraint
 - Process needs more efficient tools and workflows to address author/reviewer pain points.



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Questions

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